

Problem Set 3

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Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in **R** using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

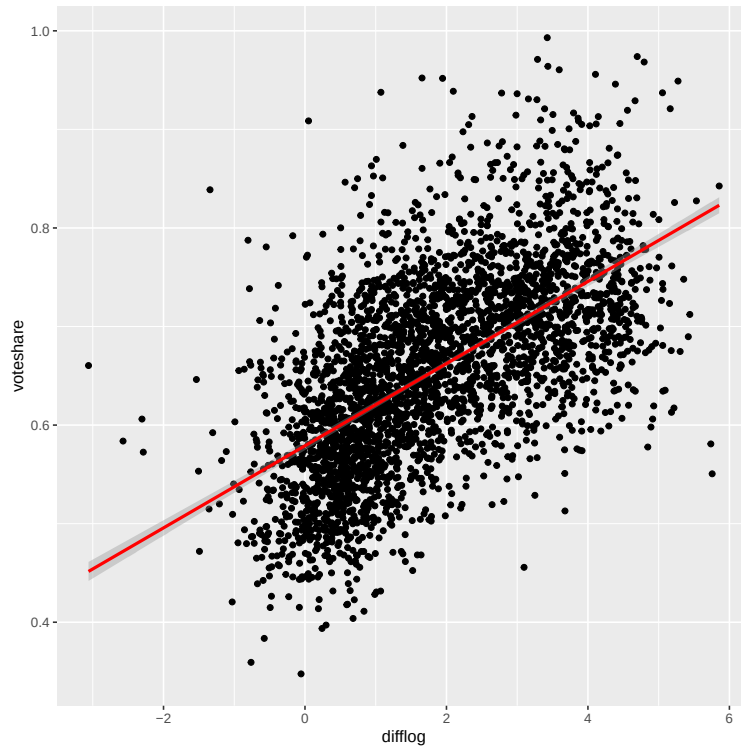
We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
1 regression1 <- lm(voteshare ~ difflog, data = inc.sub)
```

2. Make a scatterplot of the two variables and add the regression line.

```
1 scatter1 <-  
2   ggplot(inc.sub, aes(x = difflog, y = voteshare)) +  
3   geom_point() +  
4   geom_smooth(method='lm', col="red")  
5 scatter1
```



3. Save the residuals of the model in a separate object.

```
1 residuals1 <- regression1$residuals
```

4. Write the prediction equation.

```
1 regression1
```

The prediction equation is $y = 0.5790 + 0.0417x$ or $\text{voteshare} = 0.5790 + 0.0417\text{difflog}$ in the context of this specific data.

Question 2

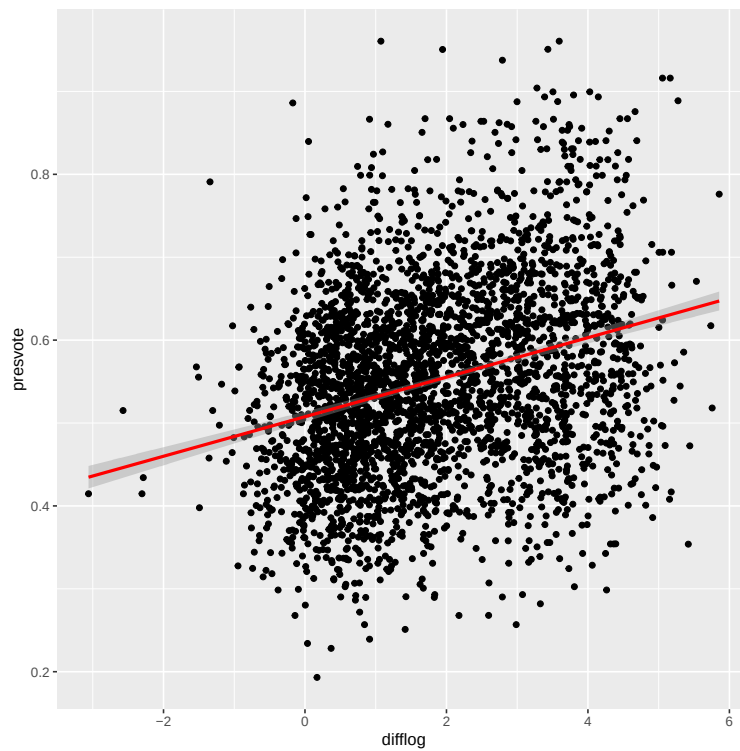
We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
1 regression2 <- lm(presvote ~ difflog, data = inc.sub)
```

2. Make a scatterplot of the two variables and add the regression line.

```
1 scatter2 <-  
2   ggplot(inc.sub, aes(x = difflog, y = presvote)) +  
3   geom_point() +  
4   geom_smooth(method = 'lm', col = "red")  
5 scatter2
```



3. Save the residuals of the model in a separate object.

```
1 residuals2 <- regression2$residuals
```

4. Write the prediction equation.

1 regression2

The prediction equation is $y = 0.5076 + 0.0238x$ or $\text{presvote} = 0.5076 + 0.0238\text{difflog}$ in the context of this specific data.

Question 3

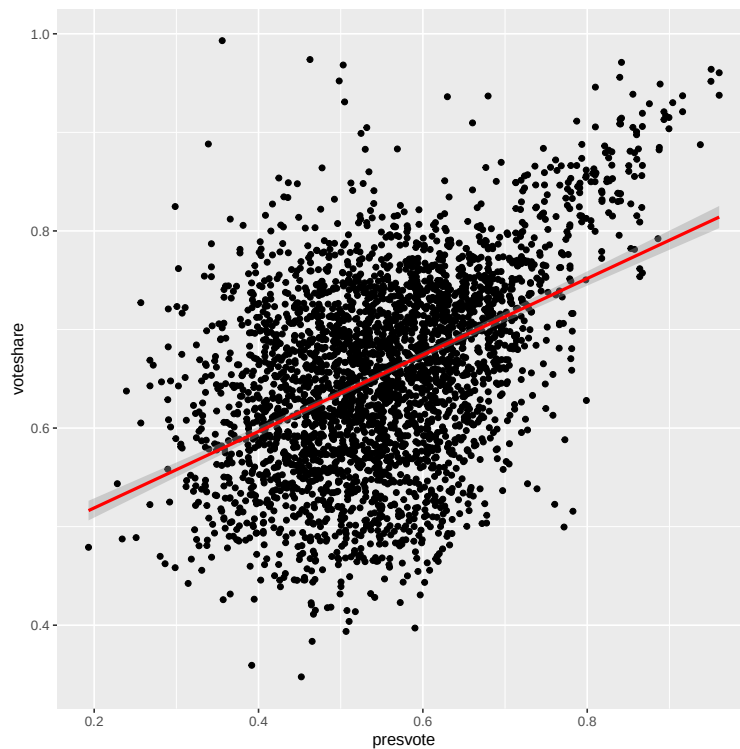
We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.

```
1 regression3 <- lm(voteshare ~ presvote, data = inc.sub)
```

2. Make a scatterplot of the two variables and add the regression line.

```
1 scatter3 <-  
2   ggplot(inc.sub, aes(x = presvote, y = voteshare)) +  
3   geom_point() +  
4   geom_smooth(method = 'lm', col = "red")  
5 scatter3
```



3. Write the prediction equation.

1 regression3

The prediction equation is $y = 0.4413 + 0.3880x$ or $\text{voteshare} = 0.4413 + 0.3880\text{presvote}$ in the context of this specific data.

Question 4

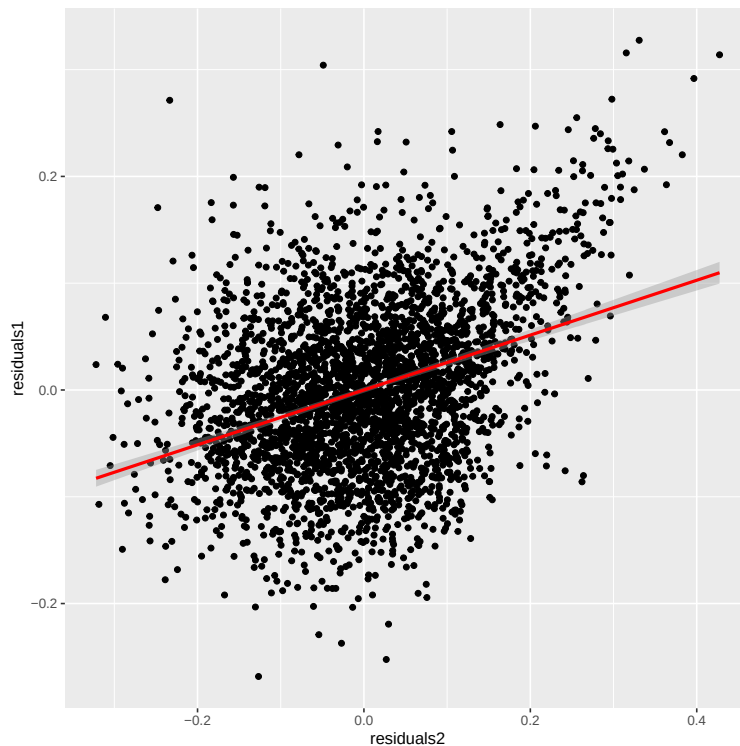
The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
1 resid_regression <- lm(residuals1 ~ residuals2)
```

2. Make a scatterplot of the two residuals and add the regression line.

```
1 df1 <- as.data.frame(cbind(residuals1, residuals2))
2 pdf("scatterplot_4.pdf")
3 scatter4 <-
4   ggplot(df1, aes(x = residuals2, y = residuals1)) +
5   geom_point() +
6   geom_smooth(method = 'lm', col = "red")
7 scatter4
```



3. Write the prediction equation.

The prediction equation is $y = -1.942\text{e-}18 + 0.2569x$ or $\text{residuals1} = -1.942\text{e-}18 + 0.2569\text{residuals2}$ in the context of this specific data.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
1 regression4 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
```

2. Write the prediction equation.

```
1 regression4
```

The prediction equation is $y = 0.4486 + 0.0355x_1 + 0.2569x_2$ or $\text{voteshare} = 0.4486 + 0.0355\text{difflog} + 0.2569\text{presvote}$ in the context of this specific regression.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

In this output and in the output in Question 4, the slope for `presvote` impact on `voteshare` is identical to the slope for `residuals2` impact on `residuals1`, which are both 0.2569. These two values are the same as they represent the same thing. `Residuals2` represents the variation in incumbent's vote share (electoral success) that is not explained by the difference in spending between incumbent and challenger, which is subsequently the variation in incumbent's vote share that is explained by the vote share of the presidential candidate of the incumbent's party, which is represented by `presvote` in the final equation. This can be concluded, as the outcome variable in the final equation is incumbent vote share and the two explanatory variables are difference in spending between incumbent and challenger and vote share of the presidential candidate of the incumbent's party. The impact of vote share of the presidential candidate of the incumbent's party is therefore equal to the variation in vote share that is not attributed to the other explanatory variable of difference in spending between incumbent and challenger.

`Voteshare` = incumbent's vote share (electoral success) `difflog` = difference in campaign spending between incumbent and challenger `presvote` = vote share of the presidential candidate of the incumbent's party `Residuals1` = variation in incumbent's vote share not explained by the difference in spending between incumbent and challenger `residuals2` = variation in the vote share of the presidential candidate of the incumbent's party not explained by the difference in spending between incumbent and challenger in the district For a one unit increase in the variation in `presvote` not explained by the difference in spending between incumbent and challenger, variation in `voteshare` that is not explained by the difference in spending between incumbent and challenger increases by 0.2569.